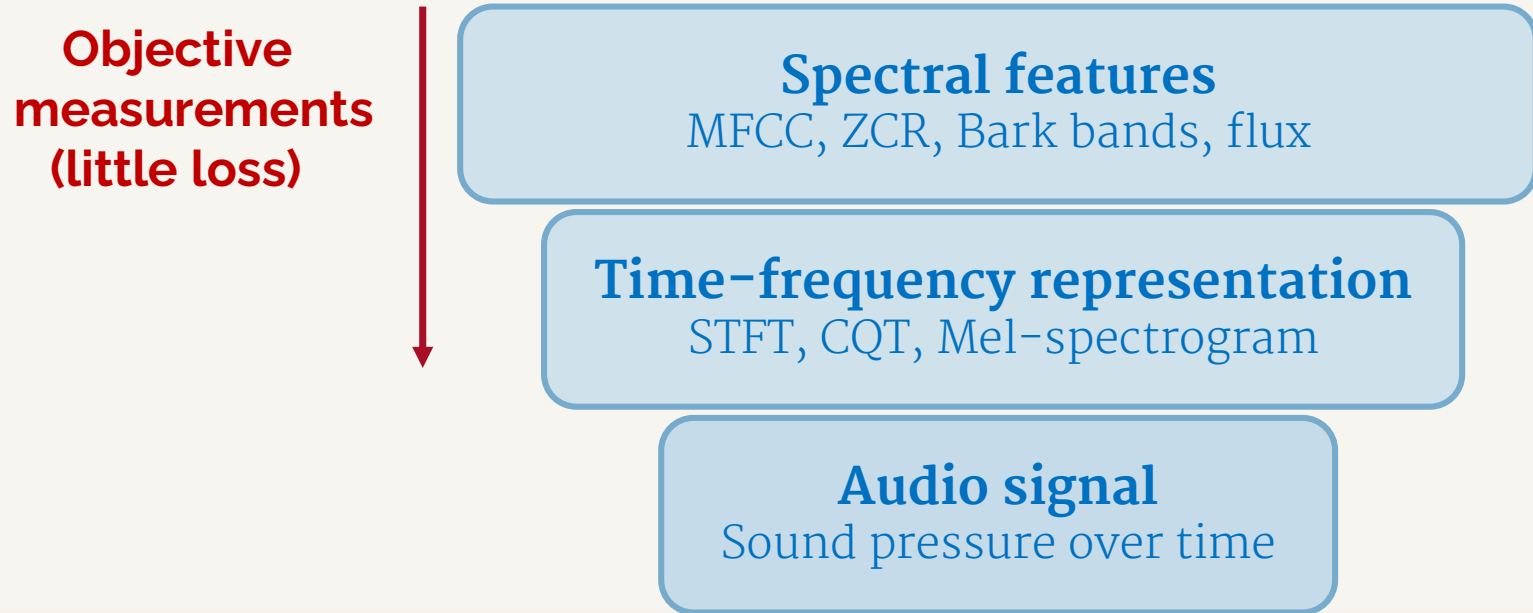


Audio Processing in Digital Musicology

Anna Aljanaki

Kunstuniversität Graz

**Objective
measurements
(little loss)**



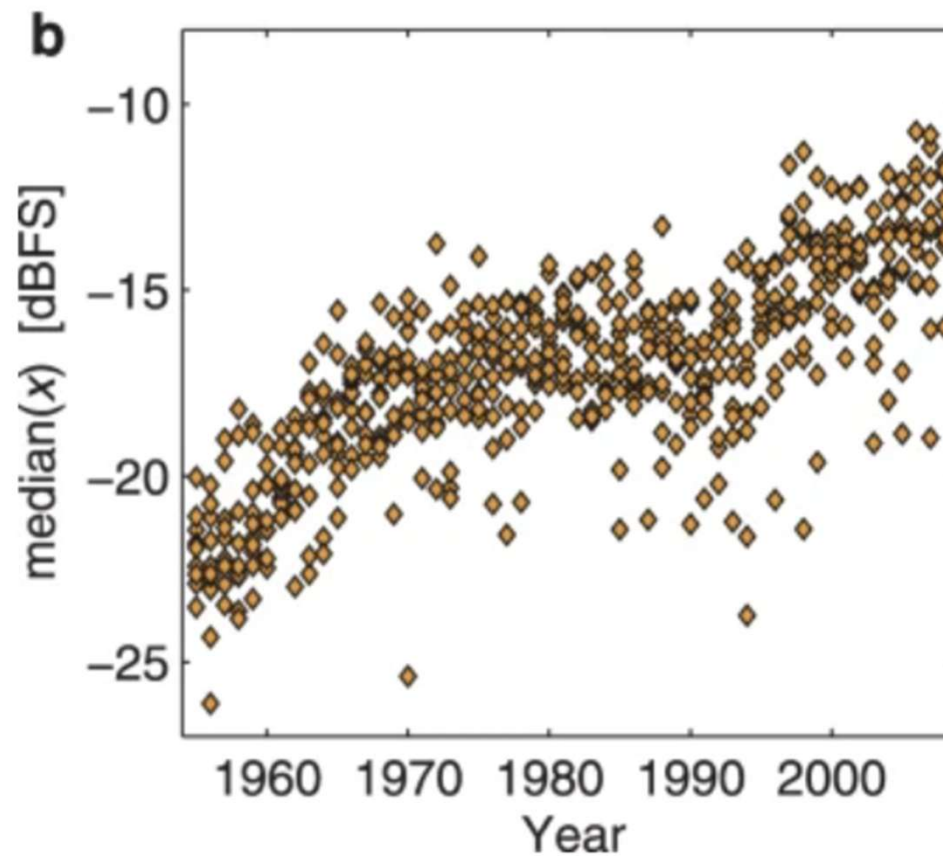
Spectral features
MFCC, ZCR, Bark bands, flux

Time-frequency representation
STFT, CQT, Mel-spectrogram

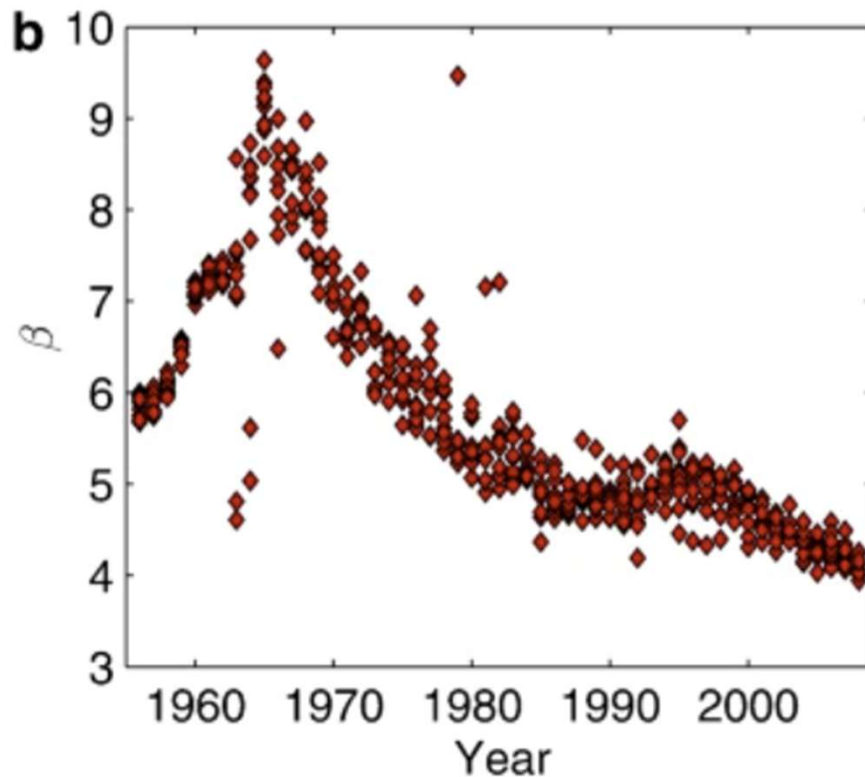
Audio signal
Sound pressure over time

Example: Evolution of Western popular music

- Approximately half a million songs
- 1955 to 2010
- Million Song Dataset
- EchoNest features: loudness, pitch and timbre



Serrà, J., Corral, Á., Boguñá, M., Haro, M., & Arcos, J. L. (2012). Measuring the evolution of contemporary Western popular music. *Scientific Reports*, 2, 521. DOI: 10.1038/srep00521



“Smaller values of β indicate less timbral variety: frequent codewords become more frequent and infrequent ones become even less frequent. This evidences a **growing homogenization of the global timbral palette**. It also points towards a progressive tendency to follow more fashionable, mainstream sonorities.”

Serrà, J., Corral, Á., Boguñá, M., Haro, M., & Arcos, J. L. (2012). Measuring the evolution of contemporary Western popular music. *Scientific Reports*, 2, 521. DOI: 10.1038/srep00521

Semantic interpretation

Emotion, tagging, similarity, recommendation

Musical structure

Source separation, segmentation, transcription, instrument detection

Musical primitives

Onsets, beats, chords

**Model-
dependent
abstractions**



Music Information Retrieval

MIR landscape



UNIVERSAL MUSIC GROUP



MIR landscape



Find me a song that
sounds similar to this

UNIVERSAL MUSIC GROUP



MIR landscape



MUSIC CREATION WITH AI

Generate me a song

UNIVERSAL MUSIC GROUP



SHAZAM

MIR landscape



What is this song
made of?



UNIVERSAL MUSIC GROUP



MIR landscape



MIR landscape



MUSIC CREATION WITH AI

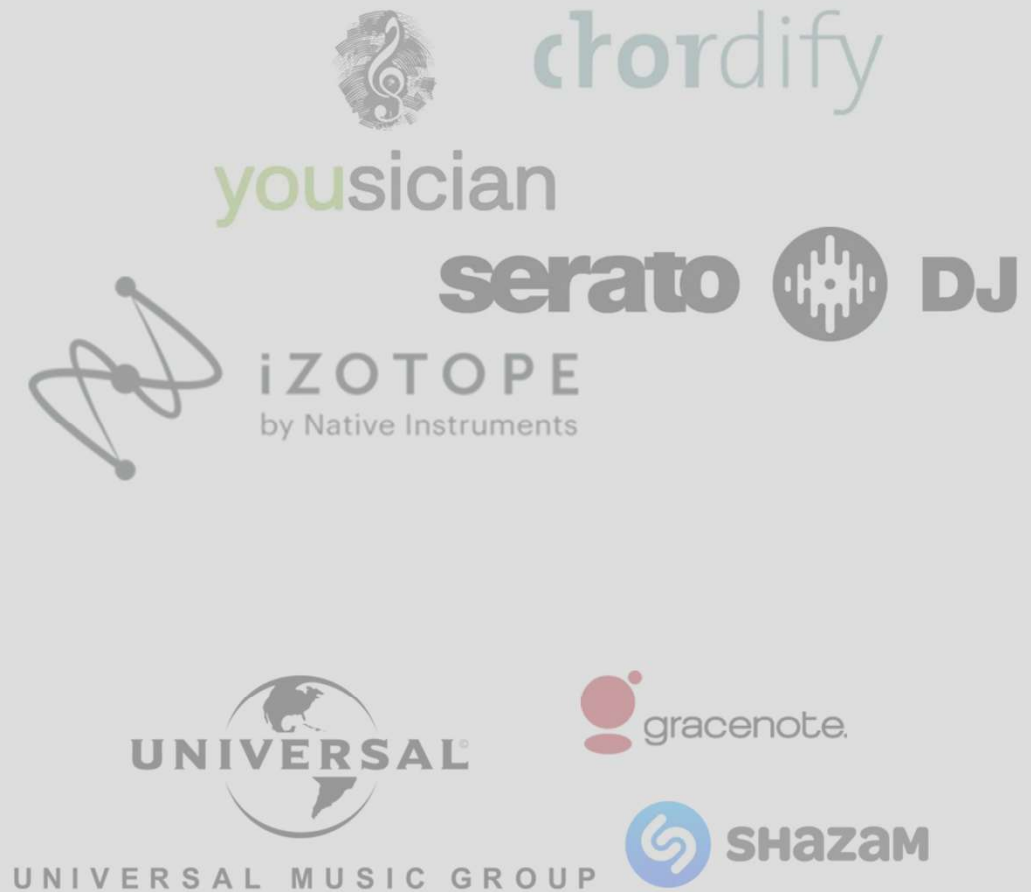


UNIVERSAL MUSIC GROUP



MIR landscape

- Audio similarity
- Music recommendation
- Categorization (mood, instrument, genre)



MIR landscape



UNIVERSAL MUSIC GROUP



MIR landscape



UNIVERSAL MUSIC GROUP



- Music generation
- Music editing
- NLP

MIR landscape



UNIVERSAL MUSIC GROUP



MIR landscape



- Chord recognition
- Melody extraction
- Transcription
- Score following
- Source separation



UNIVERSAL MUSIC GROUP



MIR landscape



UNIVERSAL MUSIC GROUP



MIR landscape



MUSIC CREATION WITH AI

- Audio fingerprinting
- AI music detection
- Anti-fraud

MIR landscape

- Music transcription
- OMR
- Musicological research



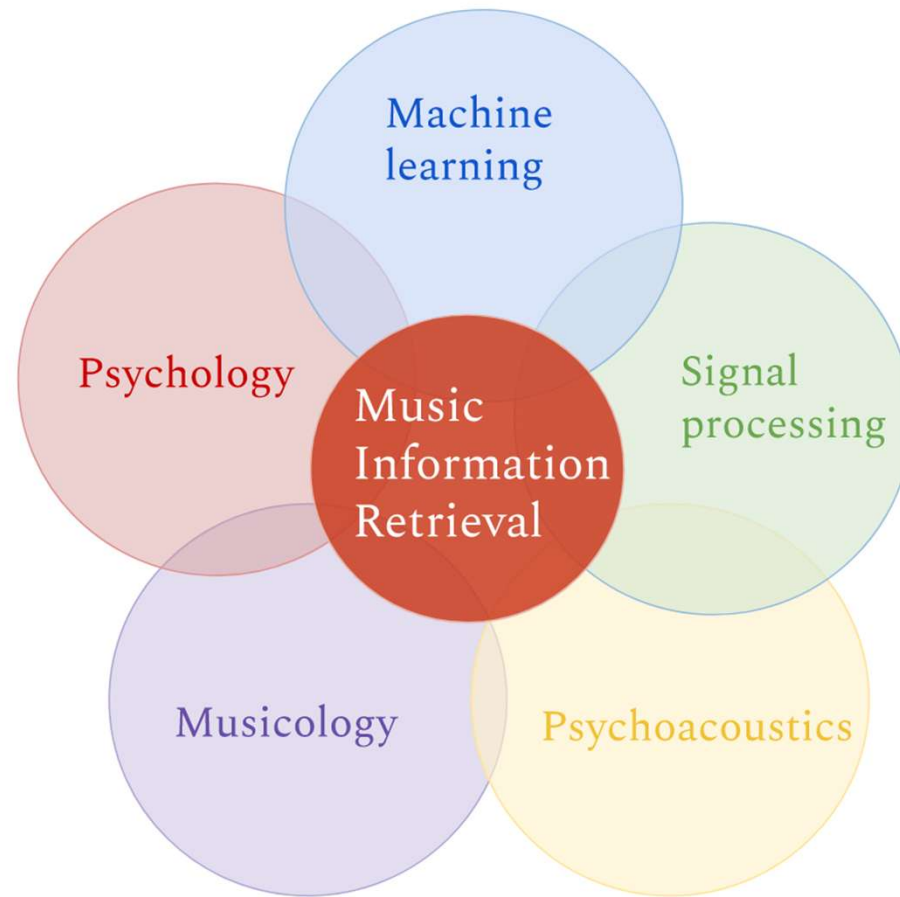
MUSIC CREATION WITH AI

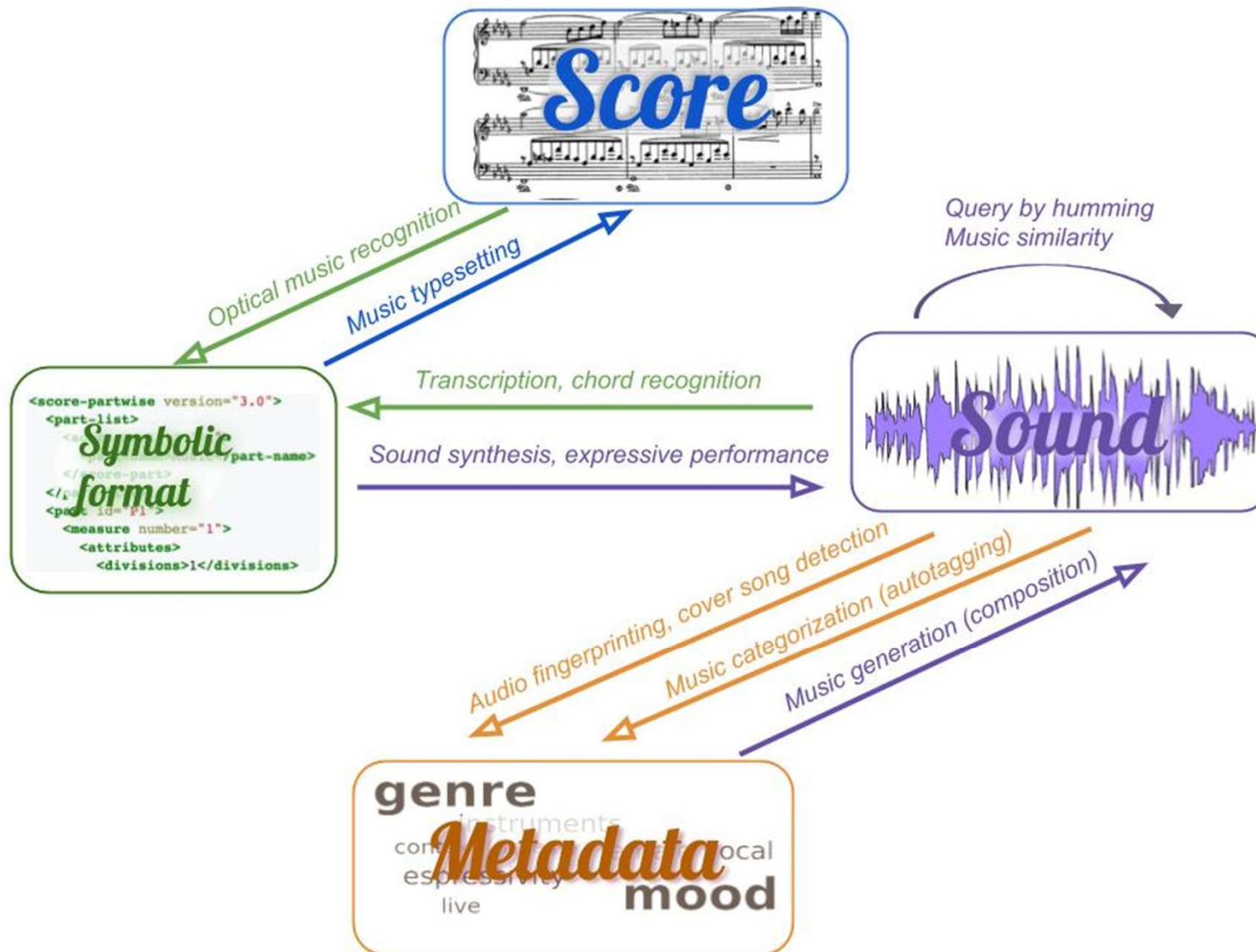


UNIVERSAL MUSIC GROUP



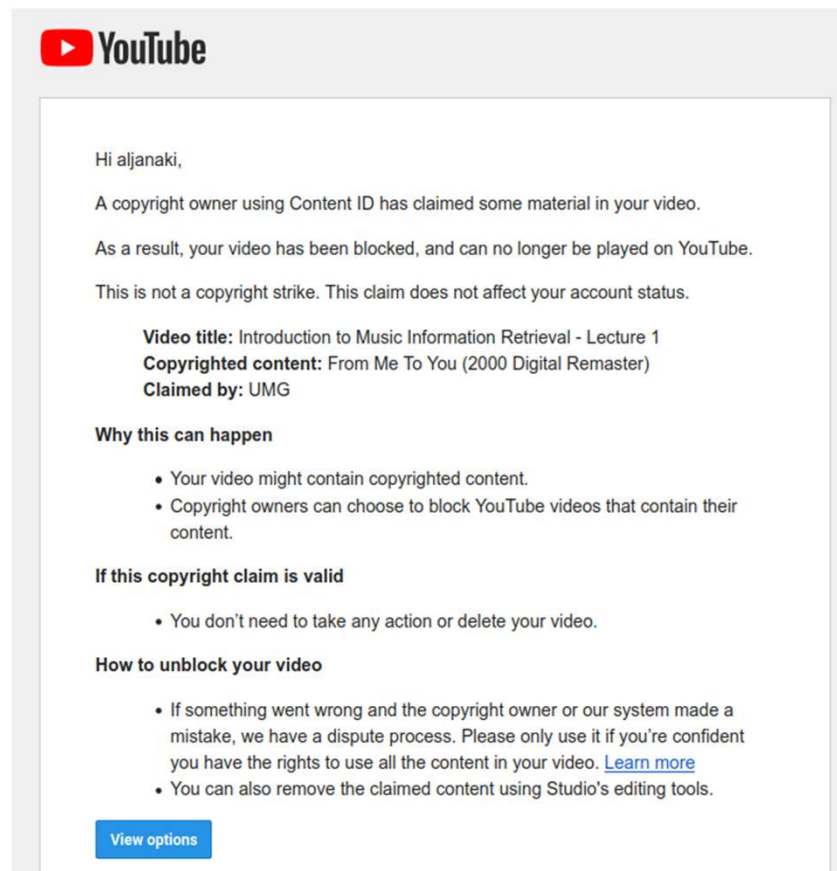
Music Information Retrieval





Audio fingerprinting

Example: audio fingerprinting



The image is a screenshot of a YouTube copyright claim notification. At the top, the YouTube logo is visible. The notification text is as follows:

Hi aljanaki,

A copyright owner using Content ID has claimed some material in your video.

As a result, your video has been blocked, and can no longer be played on YouTube.

This is not a copyright strike. This claim does not affect your account status.

Video title: Introduction to Music Information Retrieval - Lecture 1
Copyrighted content: From Me To You (2000 Digital Remaster)
Claimed by: UMG

Why this can happen

- Your video might contain copyrighted content.
- Copyright owners can choose to block YouTube videos that contain their content.

If this copyright claim is valid

- You don't need to take any action or delete your video.

How to unblock your video

- If something went wrong and the copyright owner or our system made a mistake, we have a dispute process. Please only use it if you're confident you have the rights to use all the content in your video. [Learn more](#)
- You can also remove the claimed content using Studio's editing tools.

At the bottom, there is a blue button labeled "View options".

Shazam – consumer audio fingerprinting

Try to Shazam this song:



The easily recognizable part:



Shazam – consumer audio fingerprinting

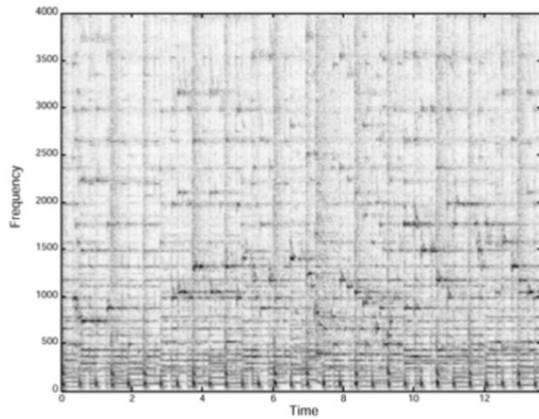


Fig. 1A - Spectrogram

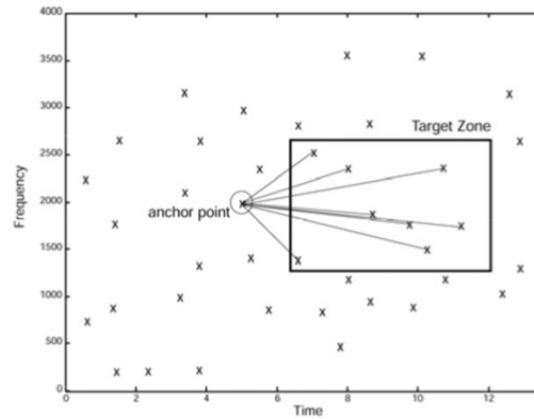


Fig. 1C - Combinatorial Hash Generation

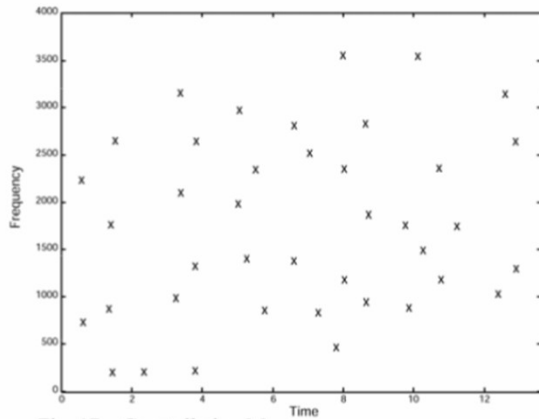


Fig. 1B - Constellation Map

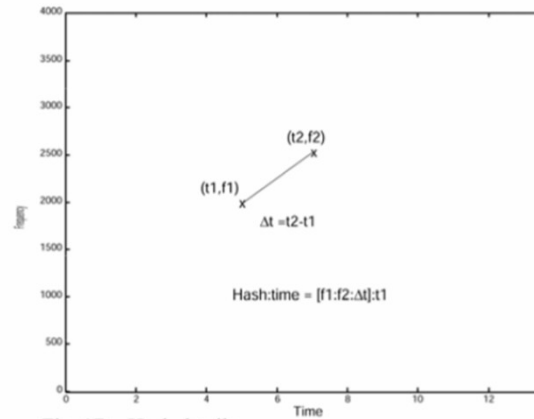
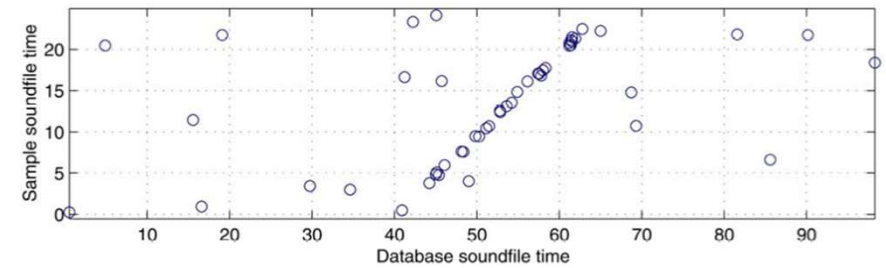


Fig. 1D - Hash details



Query by humming

- Let's try with google assistant
- Which task is more difficult for a computer to do?
Why?

Query by humming

- Extracting knowledge from a hummed or sung query:
 - Pitch
 - Rhythm
 - Phonemes
- Extracting melody from a database of songs
- Extracting similar features from the melody

How can we use this for musicology?

- Cover song detection
- Finding similar tunes in folk music archives
- Analysis of sampling in EDM
- Deduplication of field recording collections

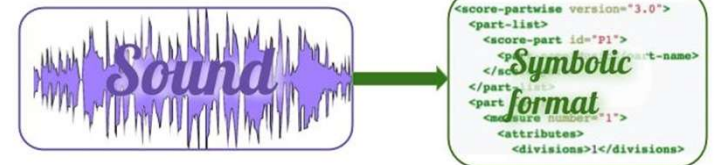
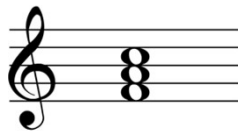
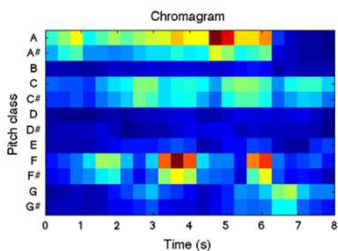
Music analysis

Tasks

- Chord recognition, key detection
- Beat tracking, onset detection
- F0 tracking (melody extraction)
- Structure segmentation
- Source separation (instrument stems)
- Lyrics transcription
- Mood recognition
- Genre recognition

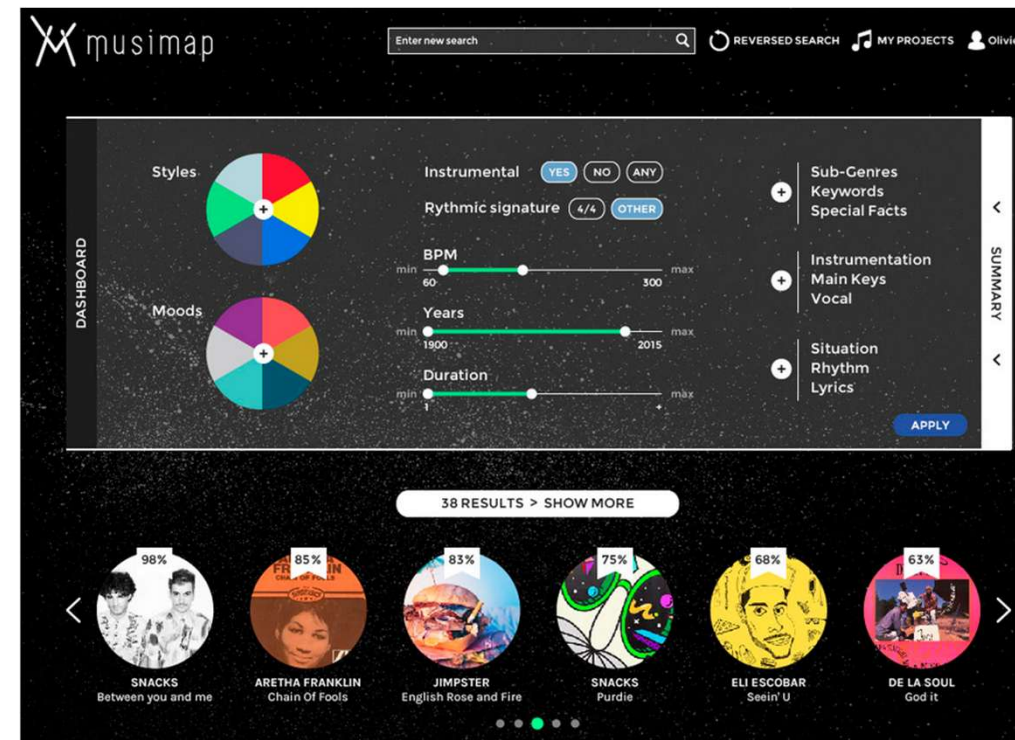
Chord recognition

- Separating harmonic and percussive information
- Extracting chord-related representation from audio (pitch heights)
- Finding the names for the chords (subjective)



Music categorization

- Feature extraction from sound
- Dimensionality reduction
- Predictive modeling
- Musimap (musimap.net)



Onset detection

- Onset – a timestamp of the beginning of the note
- Used to analyze beats

How can we use this for musicology?

- Study harmony
- Rhythm analysis
- Automatic structure analysis

Music recommendation

Similarity and recommendation

- The biggest industrial driver in MIR
- Spotify, Pandora, Youtube, TikTok, Apple Music
- Recommendation on two pillars:
 - Collaborative filtering (user listening data)
 - Content-based (music similarity)
- Content models are important for new music (cold start)

Song representation

- Traditional song representation with features
 - Spectral features
 - MFCCs
 - EchoNest features (acousticness, instrumentalness)
- Neural deep embeddings
 - MusicCNN, MERT, Jukebox, MuQ, MULE, EncodecMAE

Similar according to who?

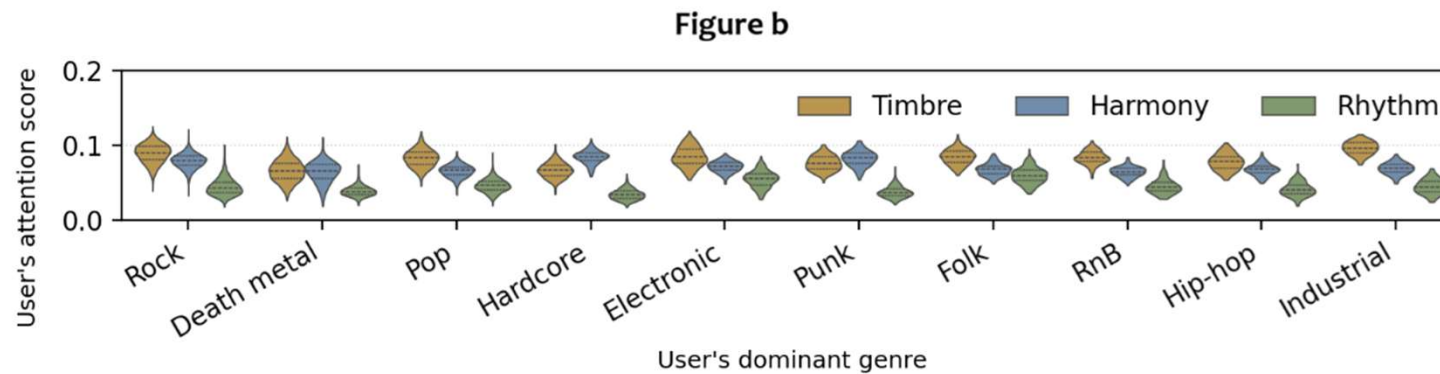
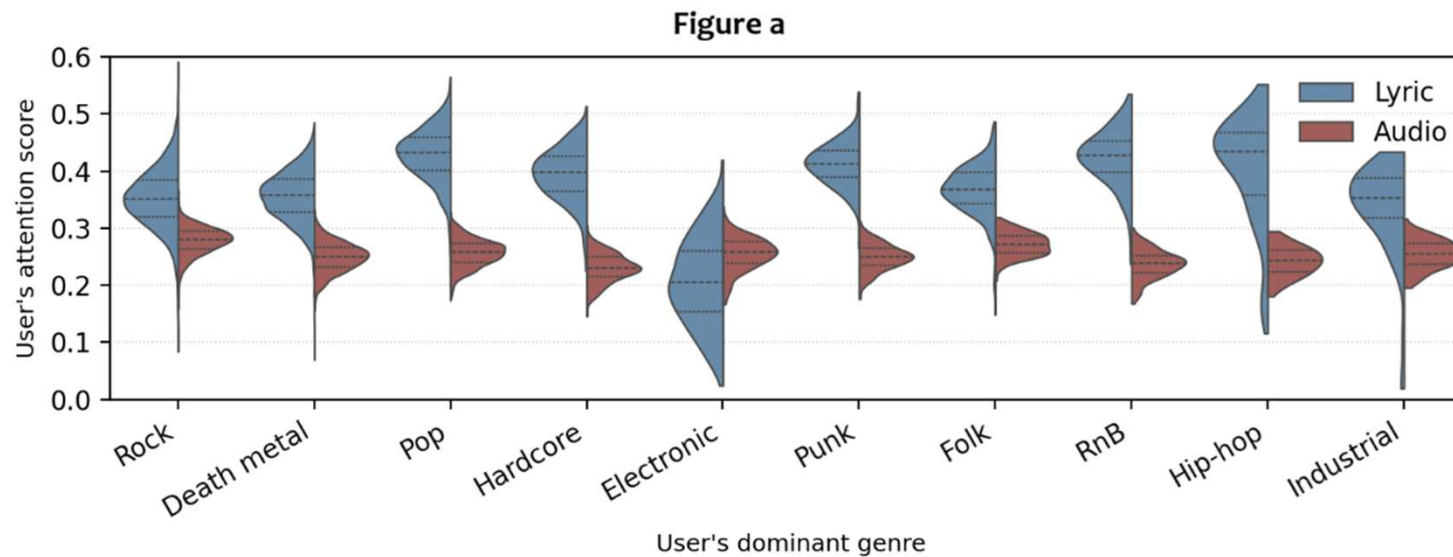
- What is the training data?
 - Spotify playlists
 - Western pop music
 - Production music
- Similarity metric
 - Music from the same sections of the same song
 - Music from the same album
 - Use multimodal data (text, video, MIDI)
 - Masked training

How can we use this for musicology?

- Song clustering by similarity
- Studying musical taste and psychology of music listening
- Cross-modal embeddings (text, MIDI, audio)

How can we use this for musicology?

- Song clustering by similarity
- Studying musical taste and psychology of music listening
- Cross-modal embeddings (text, MIDI, audio)



Yan-Martin Tamm, Anna Aljanaki, Exploring Musical Taste : Analyzing User Content Preferences through Attention Mechanisms (ISMIR 2026, under review)